13. It is given that $f(x,y)$ is a function such that

$$\frac{\partial f}{\partial x} = 2x + y^2, \quad \frac{\partial f}{\partial y} = 2xy + 3y^2,$$

and $f(2,1) = \frac{11}{2}$. Find the value of $f(1,0)$.

Give the answer as an integer or a fraction in lowest terms.

$-\frac{1}{2}$

$f_x = 2x + y^2$

Integrate wrt x,

$$f(x,y) = \int f_x \, dx = \int (2x+y^2)\, dx = x^2 + y^2 x + \phi(y) \quad (*)$$

Take $\frac{\partial}{\partial y}$ on both sides of $(*)$,

$$f_y = 0 + 2yx + \phi'(y)$$

$$\Rightarrow \quad 2xy + 3y^2 = 2yx + \phi'(y)$$

$$\Rightarrow \quad \phi'(y) = 3y^2$$

$$\Rightarrow \quad \phi(y) = \int \phi'(y)\,dy = \int 3y^2\,dy = y^3 + C$$

Substitute this into (*),

$$f(x,y) = x^2 + y^2x + y^3 + C$$

Since $f(2,1) = \frac{11}{2}$,

$$\frac{11}{2} = 2^2 + 1^2 \cdot 2 + 1^3 + C \Rightarrow \quad C = -\frac{3}{2}$$

$$\therefore \quad f(x,y) = x^2 + y^2x + y^3 - \frac{3}{2}$$

$$f(1,0) = 1 + 0 + 0 - \frac{3}{2} = -\frac{1}{2}$$

14. Let $f(x,y,z) = x^2y + y^2z + z^2x$. Determine the gradient of f at the point $(1,2,3)$.

Give the answer in the form $\langle a,b,c\rangle$, where a,b,c are real numbers.

$\langle 13, 13, 10 \rangle$

$$\nabla f(x,y,z) = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right\rangle$$

$$= \langle 2xy + z^2,\ x^2 + 2yz,\ y^2 + 2zx \rangle$$

$$\nabla f(1,2,3) = \langle 2\cdot 1\cdot 2 + 3^2,\ 1^2 + 2\cdot 2\cdot 3,\ 2^2 + 2\cdot 3\cdot 1 \rangle$$

$$= \langle 13, 13, 10 \rangle$$

Typo: 18

15. Determine the equation of the tangent plane to the surface $x^2 + y^3 + z^4 = 8$ at the point $(3, 2, 1)$.

Give the equation in the form $ax + by + cz = 1$ where each of a, b, c is an integer or a fraction in lowest terms.

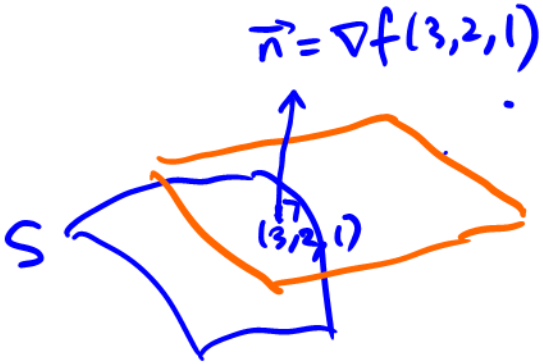
$$\frac{3}{23}x + \frac{6}{23}y + \frac{2}{23}z = 1$$

Let $f(x,y,z) = x^2 + y^3 + z^4$.

Then the given surface S is the level surface $f(x,y,z) = 18$.

$$\nabla f = \langle f_x, f_y, f_z \rangle$$

$$= \langle 2x, 3y^2, 4z^3 \rangle$$



A normal vector of the tangent plane to S at $(3,2,1)$ is given by

$$\nabla f(3,2,1) = \langle 6, 12, 4 \rangle$$

An equation of tangent plane to S at $(3,2,1)$ is:

$$\nabla f(3,2,1) \cdot \langle x-3,\ y-2,\ z-1 \rangle = 0$$

$$\Leftrightarrow \quad \langle 6,12,4 \rangle \cdot \langle x-3,\ y-2,\ z-1 \rangle = 0$$

$$\Leftrightarrow \quad 6(x-3) + 12(y-2) + 4(z-1) = 0$$

$$\Leftrightarrow \quad 6x + 12y + 4z = 46$$

$$\Leftrightarrow \quad \frac{3}{23}x + \frac{6}{23}y + \frac{2}{23}z = 1$$

16. Suppose $w = u^2 + 2uv$ where $u = xy$ and $v = -y^2$.

Determine $\dfrac{\partial w}{\partial x}$.

Give the answer as an expression in terms of x, y.

$$2xy^2 - 2y^3$$

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial w}{\partial v}\cdot\frac{\partial v}{\partial x}$$

$$= (2u+2v)\cdot y + 2u\cdot 0$$

$$= (2u+2v)y$$

$$= (2xy - 2y^2)y$$

$$= 2xy^2 - 2y^3$$

dependent variable: w

intermediate variables: u, v

independent variables: x, y

$$w = u^2 + 2uv$$

$$\Rightarrow \frac{\partial w}{\partial u} = 2u + 2v$$

$$\frac{\partial w}{\partial v} = 0 + 2u = 2u$$

$$u = xy \Rightarrow \frac{\partial u}{\partial x} = y$$

$$v = -y^2 \Rightarrow \frac{\partial v}{\partial x} = 0$$

17. Suppose $f = f(x,y,z)$ is a function of x, y, z, and x, y, z are functions of t. At $t = 0$, it is known that x, y, z all take the value 1, while their derivatives x', y', z' with respect to t all take the value 5. If $\nabla f(1,1,1) = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$, determine the value of $\dfrac{df}{dt}$ at $t = 0$.

Give the answer as an expression in terms of a, b, c.

$5a+5b+5c$

dependent variable: f

intermediate variables: x, y, z

independent variable: t

Let $r(t) = \langle x(t), y(t), z(t) \rangle$

$r'(t) = \langle x'(t), y'(t), z'(t) \rangle$

$r(0) = \langle x(0), y(0), z(0) \rangle = \langle 1,1,1 \rangle$

$r'(0) = \langle x'(0), y'(0), z'(0) \rangle = \langle 5,5,5 \rangle$.

$$\frac{df}{dt} = \nabla f(r(t)) \cdot r'(t)$$

$$\left.\frac{df}{dt}\right|_{t=0} = \nabla f(r(0)) \cdot r'(0)$$

$$= \nabla f(1,1,1) \cdot \langle 5,5,5 \rangle$$

$$= \langle a,b,c \rangle \cdot \langle 5,5,5 \rangle$$

$$= 5a + 5b + 5c$$

Alternatively,

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial f}{\partial z} \cdot \frac{dz}{dt}$$

$$\left.\frac{df}{dt}\right|_{t=0} = a \cdot 5 + b \cdot 5 + c \cdot 5$$

$$= 5a + 5b + 5c.$$

18. Consider the function $f(x,y) = a\ln(x) + xy^2$, where a is a positive constant. It is given that the directional derivative of f at the point $(1,2)$ in the direction of the vector $4\mathbf{i} - 3\mathbf{j}$ is equal to 2. Find the value of a.

$\frac{3}{2}$

$v = 4i - 3j \Rightarrow u = \frac{v}{\|v\|} = \frac{4i-3j}{\|4i-3j\|} = \frac{4i-3j}{\sqrt{4^2+(-3)^2}} = \frac{4}{5}i - \frac{3}{5}j$

$= \langle \frac{4}{5}, -\frac{3}{5} \rangle$.

$D_u f(1,2) = \nabla f(1,2) \cdot u$

$= \langle a+4, 4 \rangle \cdot \langle \frac{4}{5}, -\frac{3}{5} \rangle$

$= (a+4)\cdot\frac{4}{5} - \frac{12}{5}$

$= \frac{4}{5}a + \frac{4}{5}$

$\nabla f = \langle f_x, f_y \rangle$

$= \langle \frac{a}{x} + y^2, 2xy \rangle$

$\nabla f(1,2) = \langle a+4, 4 \rangle$

$D_u f(1,2) = 2 \Rightarrow \frac{4}{5}a + \frac{4}{5} = 2 \Rightarrow \frac{4}{5}a = \frac{6}{5} \Rightarrow a = \frac{3}{2}$.

19. Let $f(x,y) = \frac{1}{4}x^4 - \frac{1}{2}x^2 - xy + \frac{1}{2}y^2$. Find **all** critical points of $f(x,y)$.

[You do not have to classify the critical points.]

$(0,0)$, $(-\sqrt{2},-\sqrt{2})$, $(\sqrt{2},\sqrt{2})$

$f_x = x^3 - x - y$, $f_y = 0 - 0 - x + y = -x + y$

To find critical points, we solve

$$\begin{cases} f_x = 0 \\ f_y = 0 \end{cases} \Leftrightarrow \begin{cases} x^3 - x - y = 0 & -(1) \\ -x + y = 0 & -(2) \end{cases}$$

From (2), we get $y = x$. $-(3)$

Substitute (3) into (1), we get

$$x^3 - x - x = 0$$

$$\Leftrightarrow \quad x^3 - 2x = 0$$

$$\Leftrightarrow \quad x(x^2 - 2) = 0$$

$$\Leftrightarrow \quad x(x+\sqrt{2})(x-\sqrt{2}) = 0$$

$$\Leftrightarrow \quad x = 0, -\sqrt{2}, \sqrt{2}.$$

By (3),

$x = 0 \Rightarrow y = 0 \Rightarrow (x, y) = (0, 0)$

$x = -\sqrt{2} \Rightarrow y = -\sqrt{2} \Rightarrow (x, y) = (-\sqrt{2}, -\sqrt{2})$

$x = \sqrt{2} \Rightarrow y = \sqrt{2} \Rightarrow (x, y) = (\sqrt{2}, \sqrt{2})$

Check that the 3 points do satisfy (1) & (2).

20. Consider the function $f(x,y) = e^x(x^2-y^2)$. It is given that $(0,0)$ is a critical point of $f(x,y)$. Determine whether this point is a local maximum point, a local minimum point or a saddle point of $f(x,y)$.

Give the answer either as **local maximum point**, **local minimum point** or **saddle point**.

Saddle point

$f(x,y) = e^x(x^2-y^2)$

$f_x = e^x(x^2-y^2) + e^x \cdot 2x = e^x(x^2-y^2+2x)$

$f_y = -2ye^x$

$f_{xx} = (f_x)_x = e^x(x^2-y^2+2x) + e^x(2x+2)$

$= e^x(x^2-y^2+4x+2).$

$f_{xy} = (f_y)_x = -2ye^x$

$f_{yy} = (f_y)_y = -2e^x$

$$f_{xx}(0,0) = 2, \quad f_{xy}(0,0) = 0, \quad f_{yy}(0,0) = -2$$

$$D(0,0) = f_{xx}(0,0) \cdot f_{yy}(0,0) - (f_{xy}(0,0))^2$$

$$= 2 \cdot (-2) - 0^2$$

$$= -4 < 0$$

By 2nd derivative test,

$\therefore (0,0)$ is a saddle point of $f(x,y)$.

21. Let $f(x,y)$ be a function such that

$$f_x = 2x + \cos(y) \quad \text{and} \quad f_y = -x\sin(y) + 2y - \pi.$$

It is given that $f(x,y)$ has a critical point at $(0, \frac{\pi}{2})$. Determine whether this point is a local maximum point, a local minimum point or a saddle point of $f(x,y)$.

Give the answer either as **local maximum point**, **local minimum point** or **saddle point**.

local minimum point

$f_{xx} = (f_x)_x = 2 + 0 = 2$

$f_{xy} = (f_x)_y = -\sin y$

$f_{yy} = (f_y)_y = -x\cos y + 2$

$f_{xx}(0, \frac{\pi}{2}) = 2, \quad f_{xy}(0, \frac{\pi}{2}) = -1, \quad f_{yy}(0, \frac{\pi}{2}) = 2$

$$D\left(0, \frac{\pi}{2}\right) = f_{xx}\left(0, \frac{\pi}{2}\right) \cdot f_{yy}\left(0, \frac{\pi}{2}\right) - \left(f_{xy}\left(0, \frac{\pi}{2}\right)\right)^2$$

$$= 2 \cdot 2 - (-1)^2$$

$$= 3 > 0$$

$$f_{xx}\left(0, \frac{\pi}{2}\right) = 2 > 0$$

$\therefore$ By 2nd derivative test, $\left(0, \frac{\pi}{2}\right)$ is a local minimum point of $f(x,y)$.

22. Using the method of least squares, find the cubic polynomial of the form $y = ax^3 + b$ which best fits the three data points $(-1, 1)$, $(0, -7)$ and $(1, 7)$.

Give the answer in the form $y = ax^3 + b$ with the coefficients a, b in simplest form.

$y = 3x^3 + \frac{1}{3}$.

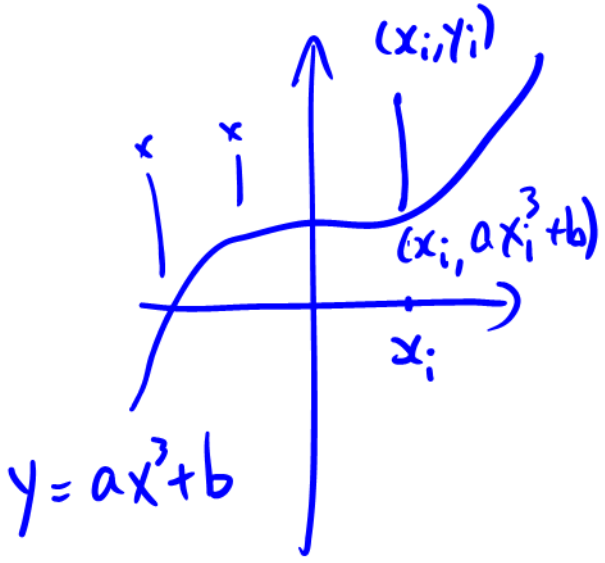
Look at the general setting of n data points (x_i, y_i) $1 \le i \le n$.

Objective: To minimize

$$f(a,b) = \sum_{i=1}^{n} (ax_i^3 + b - y_i)^2$$

$$f_a = \sum_{i=1}^{n} 2(ax_i^3 + b - y_i) \cdot x_i^3$$

$$f_b = \sum_{i=1}^{n} 2(ax_i^3 + b - y_i) \cdot 1$$



To find critical points of $f(a,b)$, we solve

$$\begin{cases} f_a(a,b)=0 \\ f_b(a,b)=0 \end{cases} \Rightarrow \begin{cases} \sum_{i=1}^{n} 2(ax_i^3+b-y_i)\cdot x_i^3 = 0 \\ \sum_{i=1}^{n} 2(ax_i^3+b-y_i)\cdot 1 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} a\sum_{i=1}^{n} x_i^6 + b\sum_{i=1}^{n} x_i^3 = \sum_{i=1}^{n} y_i x_i^3 \\ a\sum_{i=1}^{n} x_i^3 + nb = \sum_{i=1}^{n} y_i \end{cases} \quad (*)$$

$n=3$, $(-1,1)$, $(0,-7)$, $(1,7)$

$$\sum_{i=1}^{3} x_i^6 = 2, \quad \sum_{i=1}^{n} x_i^3 = -1+0+1=0, \quad \sum_{i=1}^{3} y_i x_i^3 = -1+0+7=6$$

$$\sum_{i=1}^{3} y_i = 1$$

$(*)$ becomes $\begin{cases} a\cdot 2 + 0\cdot b = 6 \\ a\cdot 0 + 3b = 1 \end{cases}$

$$\Rightarrow \begin{cases} a = 3 \\ b = \frac{1}{3}. \end{cases}$$